

# ABHISHEK VARMA

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## PROFESSIONAL SUMMARY

Computer Science and Engineering (Artificial Intelligence) student with a strong foundation in **data structures, algorithms, full-stack development, and AI engineering**. Hands-on experience building AI-powered applications using **LLMs, RAG, agentic workflows, vector databases, and modern web technologies**. Experienced with Java, TypeScript, JavaScript, Python, React, Node.js, FastAPI, PostgreSQL, MongoDB, Redis, LangChain, and LangGraph. Currently focused on building practical AI and software engineering projects.

## EDUCATIONS

**Muthoot Institute of Technology and Science**|Kochi, India  
*Bachelor of Technology in Computer Engineering*

2024 - 2028  
CGPA : 8.09/10

## PROJECTS

**NexoraAI—Multi-Agent AI Platform** | 🐙 Github 2026 - Present

- Building a **multi-agent AI platform** using MERN stack and a scalable **microservices architecture**.
- Developed separate **API Gateway, authentication, chat, and agent services** using Node.js and Express.js.
- Implemented **JWT authentication, authentication middleware, Redis-based session management**, and current-user APIs.
- Built AI workflows using **LangChain and LangGraph**, including agent APIs, tool calling, and multi-step workflows.
- Integrated **RAG, Qdrant vector search, embeddings, and Redis-based AI memory** for context-aware AI interactions.
- Developed the frontend using **React and Redux Toolkit**, including authentication, chat interface, sidebar, and chat-area components.

*Technologies / Tools Used:* MERN Stack, LangChain, LangGraph, Qdrant, Redis, Docker

**CommercePilot – AI-Driven Agentic Commerce Platform** | 🐙 Github 2026 - Present

- Engineered an autonomous AI buyer agent that parses natural language queries, evaluates product specifications against user budget constraints, and proposes structured checkout carts.
- Architected a **RAG-based search engine** using PostgreSQL pgvector, replacing rigid keyword matching with **1536-dimensional semantic vector retrieval** for merchant catalogs.
- Developed a secure transactional backend using Node.js, Express, and TypeScript to manage merchant onboarding, **ACID-compliant inventory updates**, and order lifecycles.
- Implemented enterprise-grade AI safety guardrails, capturing **100%** of LLM tool-calling logic, retrieved context, and product reasoning into an immutable database audit trail.
- Integrated OpenAI/Gemini function calling with a gated Razorpay checkout flow, ensuring that all AI-initiated financial transactions require explicit user approval and handle failures gracefully.
- Containerized the end-to-end application stack—including the API server, pgvector database, and Redis cache—using Docker Compose for reliable state management and seamless local deployment.

*Technologies / Tools Used:* Node.js, TypeScript, Express.js, PostgreSQL, pgvector, Redis, React, Vite, Docker, OpenAI/Gemini, LangGraph, Razorpay API

**PageIntel | AI Web Assistant Chrome Extension** | 🐙 Github 2026 - Present

- Developed a Chrome extension and FastAPI backend enabling users to chat with the content of any webpage using 10 different LLMs (GPT-4o, Claude, DeepSeek).
- Built a **RAG** pipeline with Qdrant vector database and LangChain to chunk and embed page content for context-aware answers.
- Implemented an automatic fallback to DuckDuckGo web search for missing information and built a side-by-side model comparison mode.
- Designed persistent, domain-specific memory using Redis to store and recall chat history dynamically based on the active URL.

*Technologies / Tools Used:* Python, FastAPI, Qdrant, Redis, JavaScript

## SKILLS

|                                 |                                                                             |
|---------------------------------|-----------------------------------------------------------------------------|
| <b>Programming Languages:</b>   | Java, Python, JavaScript, TypeScript                                        |
| <b>Frontend :</b>               | React.js, Tailwind CSS, Next.js, Vite                                       |
| <b>Backend :</b>                | Node.js, Express.js, Fast API, REST APIs                                    |
| <b>Generative AI &amp; LLM:</b> | RAG Pipelines, LangChain, LangGraph, LLMs, Vector Database, LLM Integration |
| <b>Tools &amp; Platforms:</b>   | Git, GitHub, Docker, Postman, VS Code                                       |

## PROBLEM SOLVING

💻 LeetCode

**Problem Solving:** Solved **200+ Data Structures & Algorithms** problems on LeetCode, covering arrays, hashing, trees, graphs, recursion, binary search, and dynamic programming